



A word cloud graphic on a white background, partially enclosed by a dark grey curved shape. The words are in various shades of green and brown. The most prominent word is 'Power' in large, bold, dark green letters. Other words include 'estimate' in brown, 'formula' in green, 'error' in green, 'sample' in brown, 'Size' in green, and 'size' in green.

Power Analysis

11/06/2025 (Week 11)

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Outline

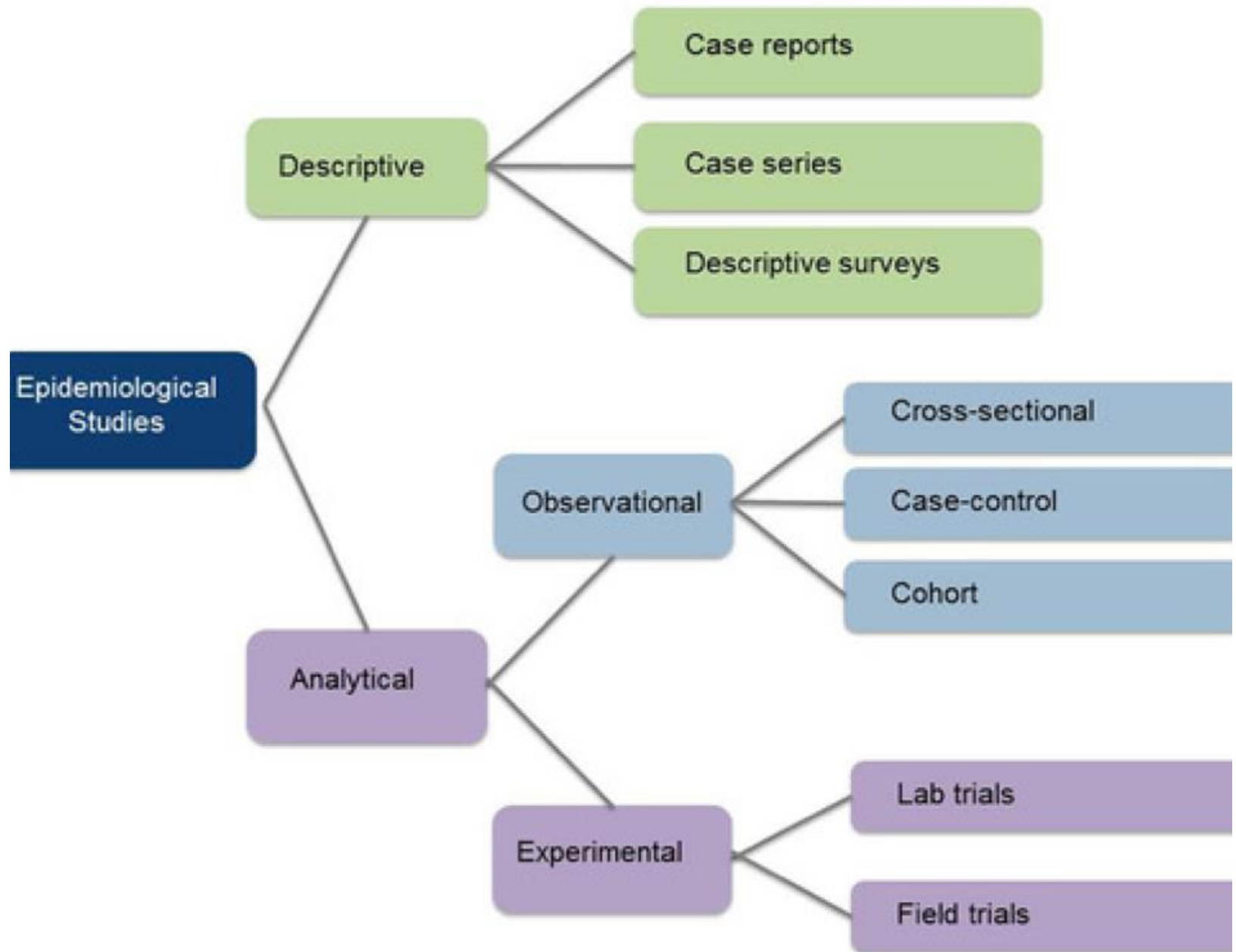
Study Design

Review Type I and Type II Error

Power

Power Analysis

Study Design





1. Observational Studies

Researchers **observe** subjects without intervening or manipulating variables.

Type	Description	Example
Cross-sectional	Data collected at one point in time to describe or find associations.	Measuring prevalence of diabetes in adults today.
Cohort	Follows a group over time to see who develops an outcome.	Tracking smokers and non-smokers for 10 years to see who develops lung cancer.
Case-control	Compares people with a condition (cases) to those without (controls) to look for past exposures.	Comparing patients with lung cancer vs. healthy controls to see past smoking habits.
Ecological	Looks at group-level data rather than individual-level data.	Comparing countries' average sugar consumption and diabetes rates.

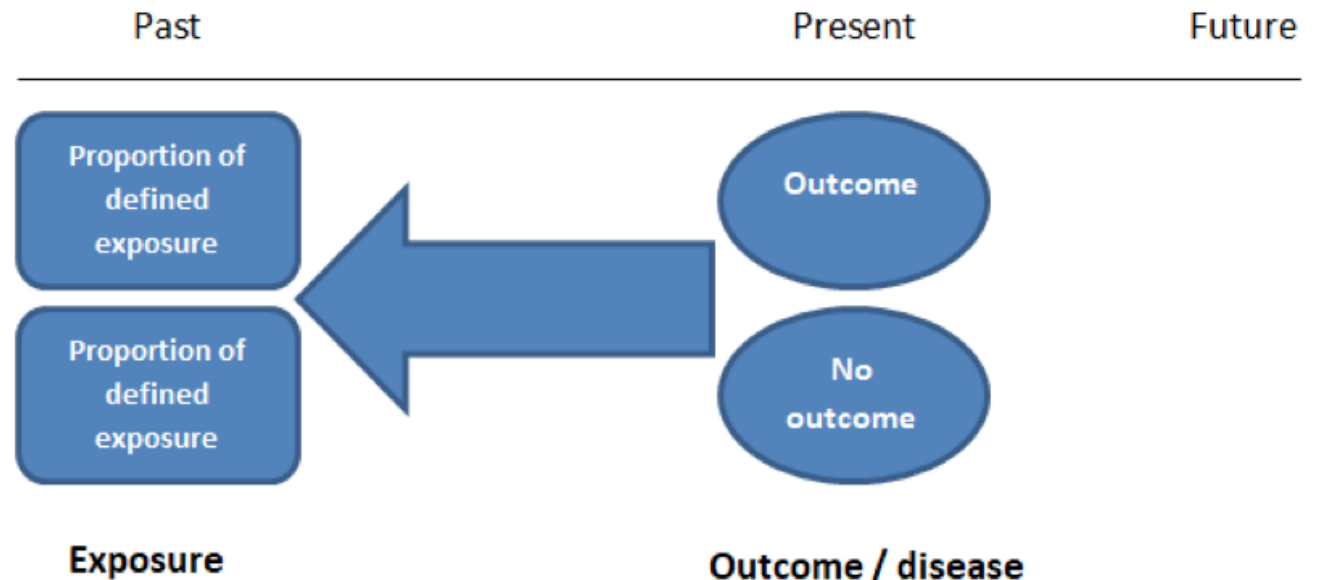
Case-Control Study

- We conduct this study by comparing 2 groups: one group with the disease (**cases**) and another group without the disease (**controls**)
- This design is always **Retrospective**
- We aim to find out the odds of having a risk factor or an exposure if an individual has a specific disease (**Odds ratio**)
- Relatively easy to conduct

Case-Control Study

Example:

- We want to study the odds of being a smoker among hypertensive patients compared to normotensive ones.
- To do so, we choose a group of patients diagnosed with hypertension and another group that serves as the control (normal blood pressure).
- Then we study their smoking history to find out if there is a correlation.



Study Design

2. Experimental (Interventional) Studies

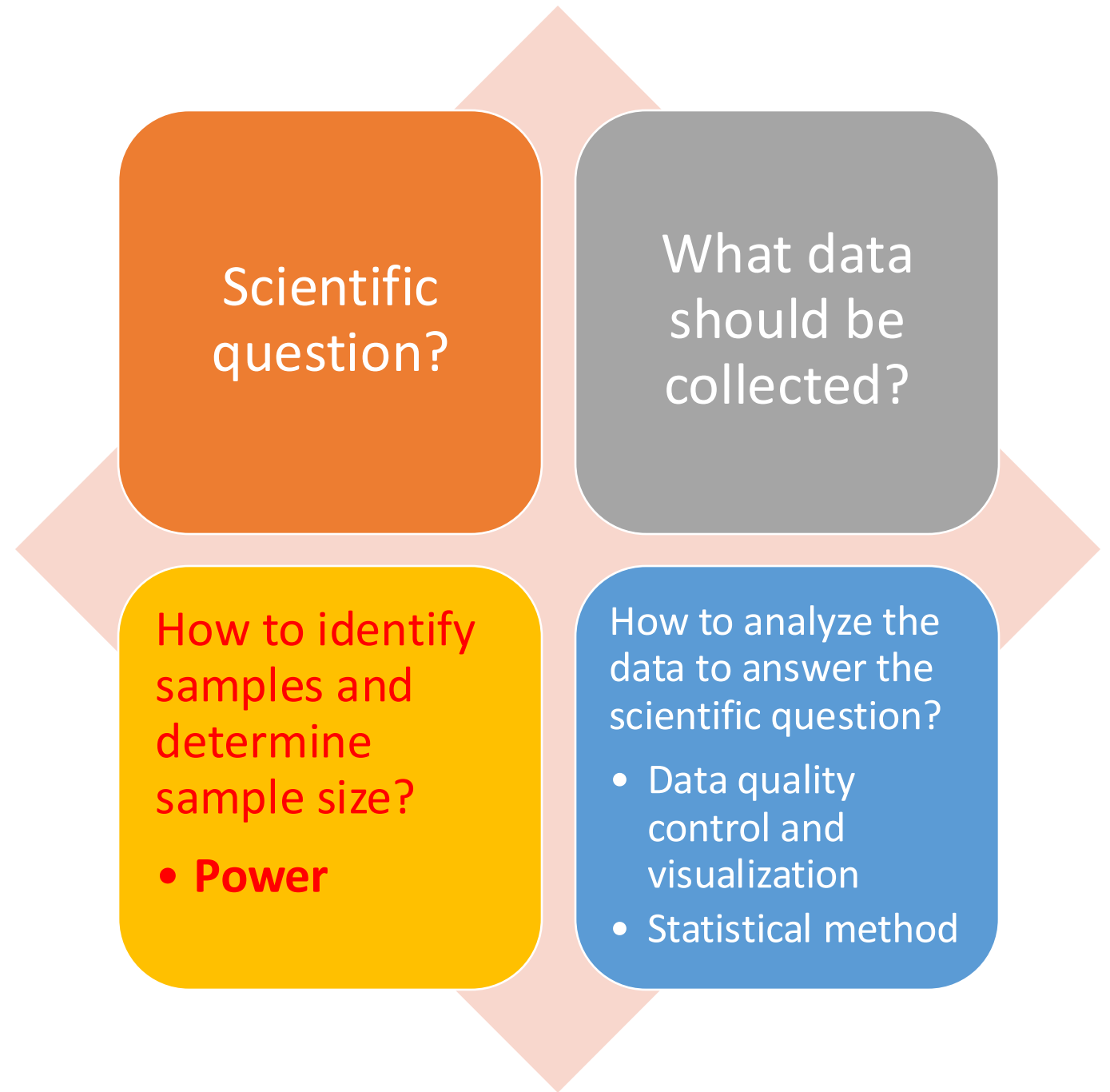
Researchers **actively intervene** — for example, assigning a treatment — to see its effect.

Type	Description	Example
Randomized Controlled Trial (RCT)	Participants are randomly assigned to intervention or control groups.	Testing a new drug vs. placebo.
Quasi-experimental	Intervention is applied but without full randomization.	Implementing a new school program in one district but not another.

Experimental Studies

- Also known as interventional studies
- Can involve animals and humans. Pre-clinical trials involve animals
- Clinical trials are experimental studies involving humans
- In randomized controlled trials, one group of participants receives the control, while the other receives the tested drug/intervention. Those studies are the best way to evaluate the efficacy of a treatment.

Questions Need to Address?



How important is statistical power?

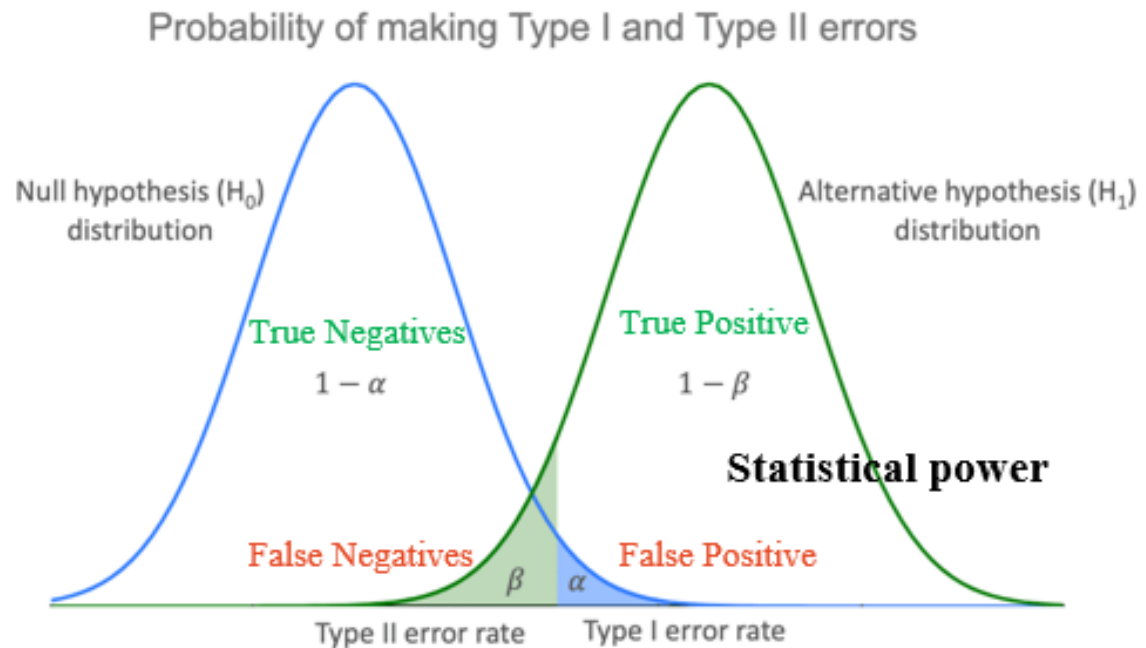
- Important for funding
 - Needed for a successful grant application: Rigor
 - NIH applications often rejected due to:
 - Low power (sample size too small)
 - Lack of power calculations
 - Inappropriate power calculations
- Important for success of study
 - Underpowered studies are less likely to yield significant findings
 - When they do yield significant results, much more likely to be false positives
- Need to consider in every study: validity, power, type I error

Hypothesis Testing

- Null Hypothesis: H_0
 - Alternative Hypothesis: H_a
 - Test Statistic: T
1. Determine the distribution (probability density function) of T under H_0
 2. Calculate the value of test statistic T_{obs} with observed data
 3. Calculate p-value under H_0 :
 $Prob(T \text{ is as or more extreme than } T_{obs} | H_0)$
 4. Reject H_0 if p-value $< \alpha$.

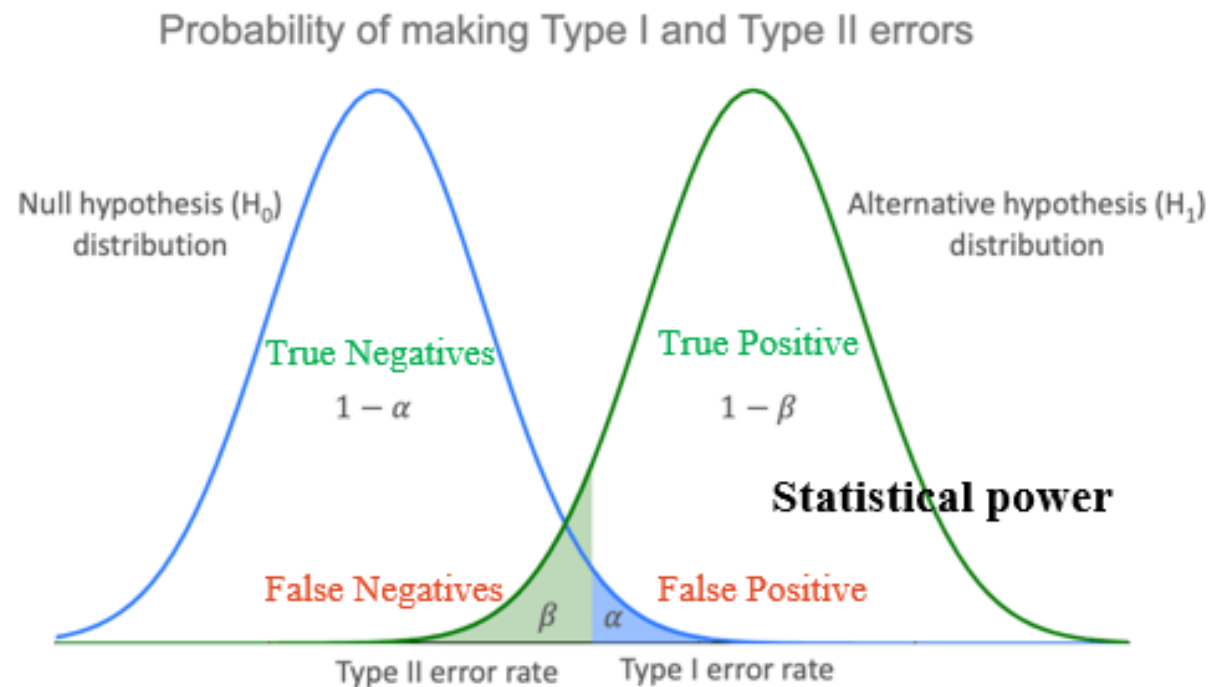
Type I and Type II Error

	H_0 is TRUE	H_0 is FALSE
Not Reject H_0	Correct Inference (true negative) (probability = $1 - \alpha$)	Type II error (false negative) (probability = β)
Reject H_0	Type I error (false positive) (probability = α)	Correct Inference (true positive/Power) (probability = $1 - \beta$)



Which Test Statistic/Method to Use?

- Dose lower Type I Error mean lower Type II Error as well?
- How to control for Type I Error?
- How to improve power?
- Should be appropriate for your data type!



Power



Power

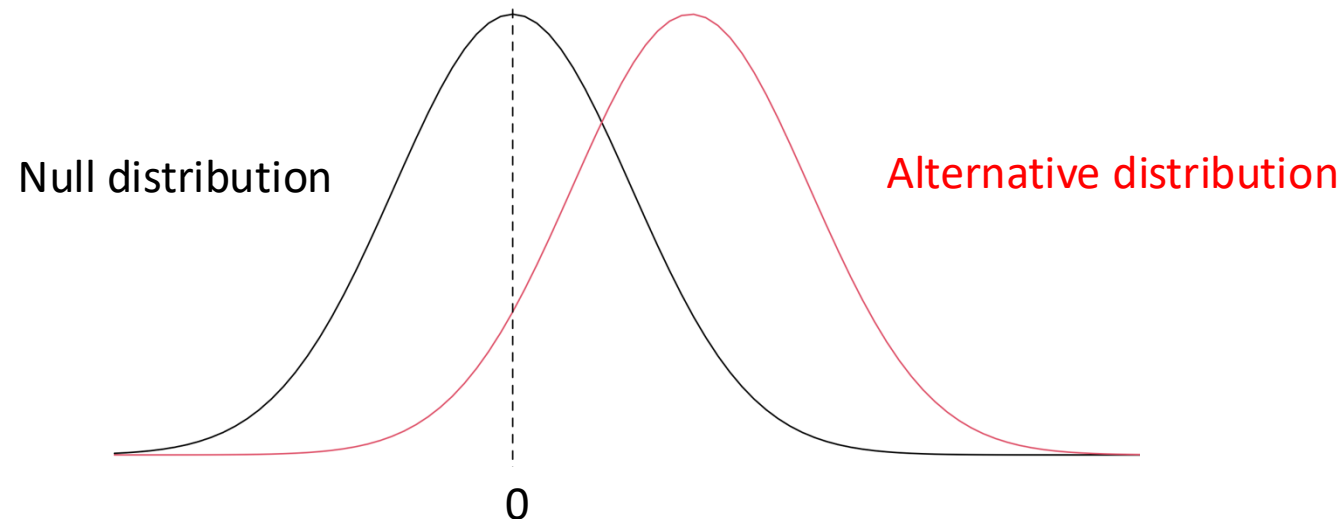
The Probability to Reject Null Hypothesis
when the Null Hypothesis is Truly FALSE
(True Positive Probability)

$$\text{Power} = 1 - \text{Type II Error } (\beta)$$

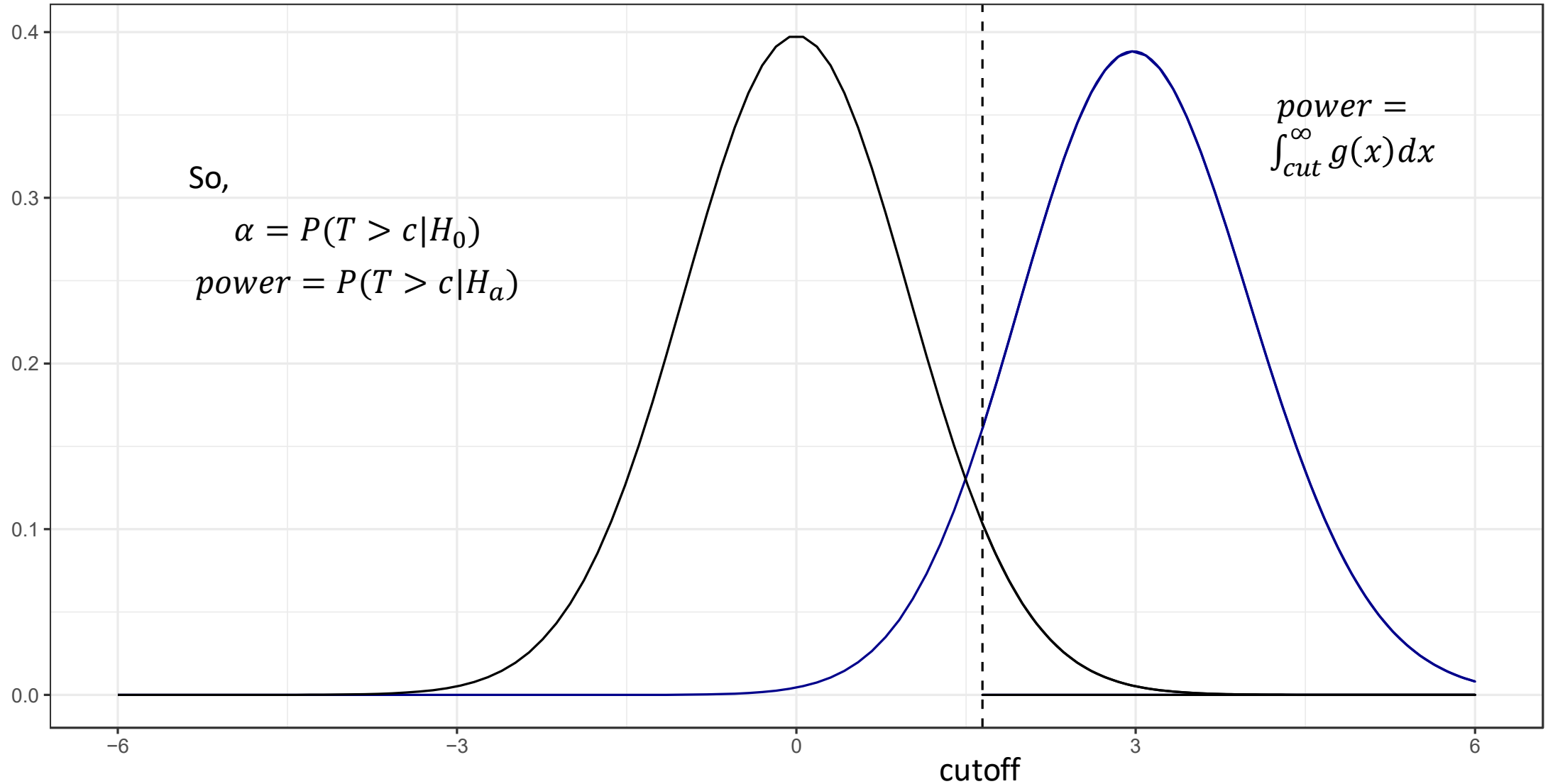
- Chance of detecting a real/true effect
- Analytical power function would be known if we know the probability density distribution of test statistic under H_a

Alternate distribution

- T-test example: $T = \frac{\bar{X}_1 - \bar{X}_2}{s/\sqrt{n}}$
 - s is standard deviation of outcome data
 - n is sample size
- How is T distributed under the alternative hypothesis?
- Alternative hypothesis is that $\bar{X}_1 - \bar{X}_2 > 0$. But this is not specific!
- If we **assume a particular effect size**, it is possible to derive the distribution of T *assuming that effect size is correct*.

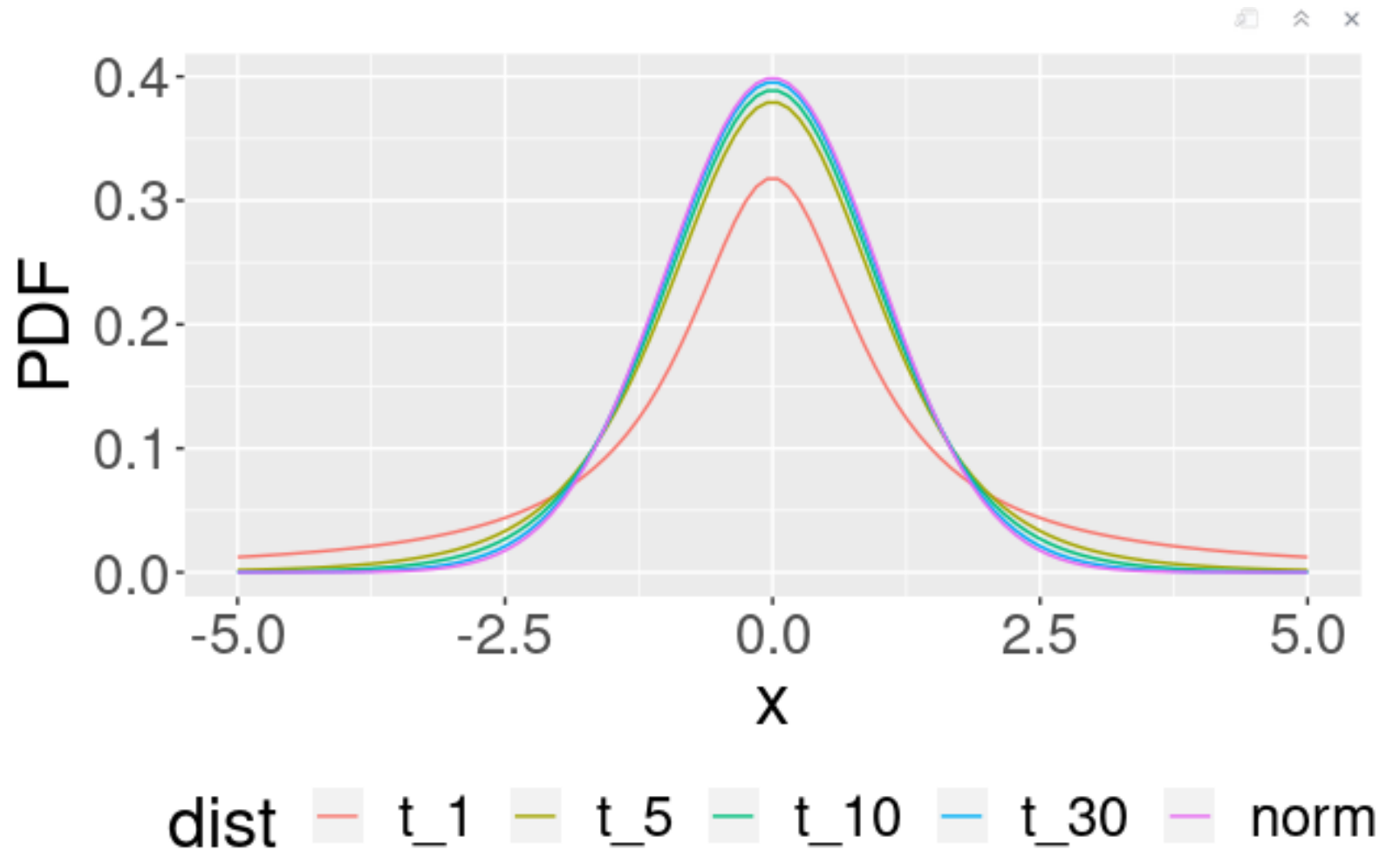


Power can be computed as AUC from alternative distribution



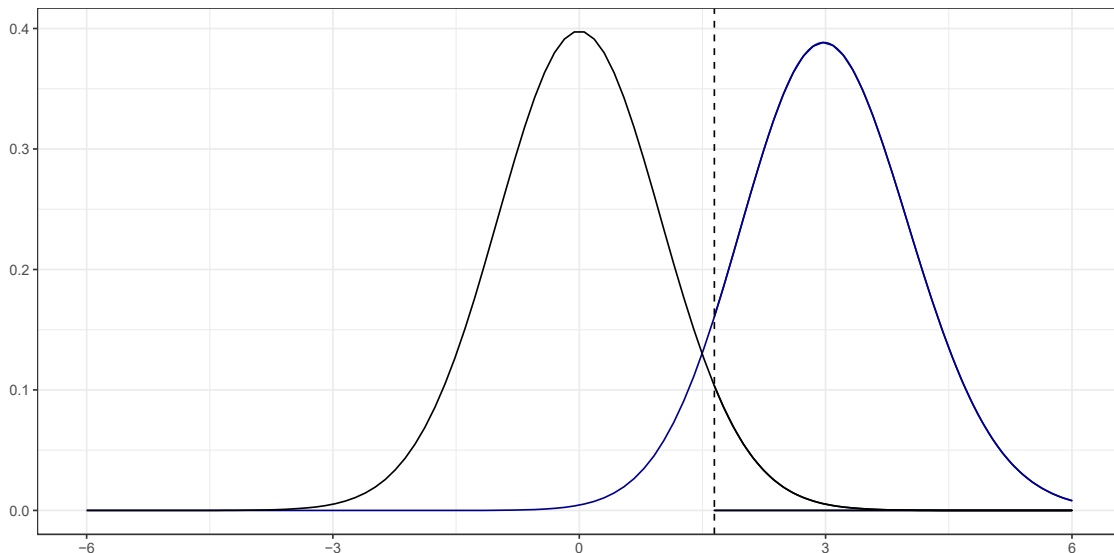
T-distribution and the standard normal distribution

T-distribution would converge to a standard normal distribution as the degrees of freedom increase!



What factors influence the power of a study?

1. Sample size
2. True effect size (size of difference or association in the population)
 - Mean difference
 - Slope
 - Correlation r or r^2
3. Significance criterion – what α -level are we using?



- **The alternate distribution depends on true effect size and sample size**
- **The significance criterion determines how easy/hard it is to reject H_0**

A General Formula for Power

$$Power = 1 - \beta \propto \frac{ES \alpha \sqrt{n}}{\sigma}$$

- ES : Effect Size
- α : Type I Error
- n : Sample Size
- σ : Outcome Standard Deviation

$$n \propto \left(\frac{\sigma (1 - \beta)}{\alpha ES} \right)^2$$

$$ES \propto \frac{(1 - \beta)\sigma}{\alpha \sqrt{n}}$$

Effect Size (ES)

Type of Test	Effect Size (ES)
One-sample t-test	Sample mean
Two-sample t-test	Differences between means
Linear regression	Slope/coefficient
One-way ANOVA	Differences between means

Review of effect sizes

Effect size (useful for interpretation)	Test statistic (useful for inference)
Estimate of the underlying “true value”	Variance-standardized version of this estimate
Scale may vary, may depend on units of measurement	Standardization solves this problem
Example: group difference in means $\bar{X}_1 - \bar{X}_2$	Example: T-statistic $\frac{\bar{X}_1 - \bar{X}_2}{s/\sqrt{n}}$
Other examples: correlation coefficient, regression slope, odds ratio, Cohen’s d (Cohen’s d = $(\bar{X}_1 - \bar{X}_2)/s$)	Other examples: F-statistic, chi-squared statistic, p-value
Does not depend on N under null or alternative	If alternative is true, becomes more extreme as N increases (why?)

When Power Function is Analytically Known

R library “pwr”

- `pwr.t.test()` for t-test

For t-tests, the effect size is assessed as

$$d = \frac{|\mu_1 - \mu_2|}{\sigma}$$

where μ_1 = mean of group 1
 μ_2 = mean of group 2
 σ^2 = common error variance

Cohen suggests that d values of 0.2, 0.5, and 0.8 represent small, medium, and large effect sizes respectively.

Details

Exactly one of the parameters 'd', 'n', 'power' and 'sig.level' must be passed as NULL, and that parameter is determined from the others. Notice that the last one has non-NULL default so NULL must be explicitly passed if you want to compute it.

Input/Output Arguments for `pwr.t.test()`:

- **n** : Sample Size
- **d** : Cohen’s Effect Size (standardized), difference between the means divided by the pooled standard deviation
- **power** : Power
- **sig.level**: significance level, e.g., 0.05 by default
- **type** : type of test, e.g., "two.sample", "one.sample", "paired"
- **alternative** : “two.sided”, “less”, “greater”

Power Analysis



Example 1: One-sample t-test

Example 1: Determine the power For given sample size n and effect size d

```
### Find the power for given sample size and effect size
```

```
```{r}  
pwr.t.test(n = 100, d = 0.29, sig.level = 0.05,
 type = "one.sample", alternative = "two.sided")
```
```

One-sample t test power calculation

```
      n = 100  
      d = 0.29  
sig.level = 0.05  
power = 0.8190623  
alternative = two.sided
```

Example 1: One simulation with sample size 100 and mean (effect size) 0.29

```
```{r}
n = 100
Simulate x variable values
x = rnorm(n, mean = 0.29, sd = 1)
t.test(x)
```
```

One Sample t-test

```
data: x
t = 3.3162, df = 99, p-value = 0.001276
alternative hypothesis: true mean is not equal to 0
95 percent confidence interval:
 0.1297010 0.5161124
sample estimates:
mean of x
0.3229067
```

```
```{r}
sample mean
x_mean = mean(x)

sample deviation
x_sd = sd(x)

t-test statistic value
t_stat = x_mean / (x_sd / sqrt(n))

print(c(x_mean, x_sd, t_stat))
```
```

```
[1] 0.3229067 0.9737125 3.3162425
```

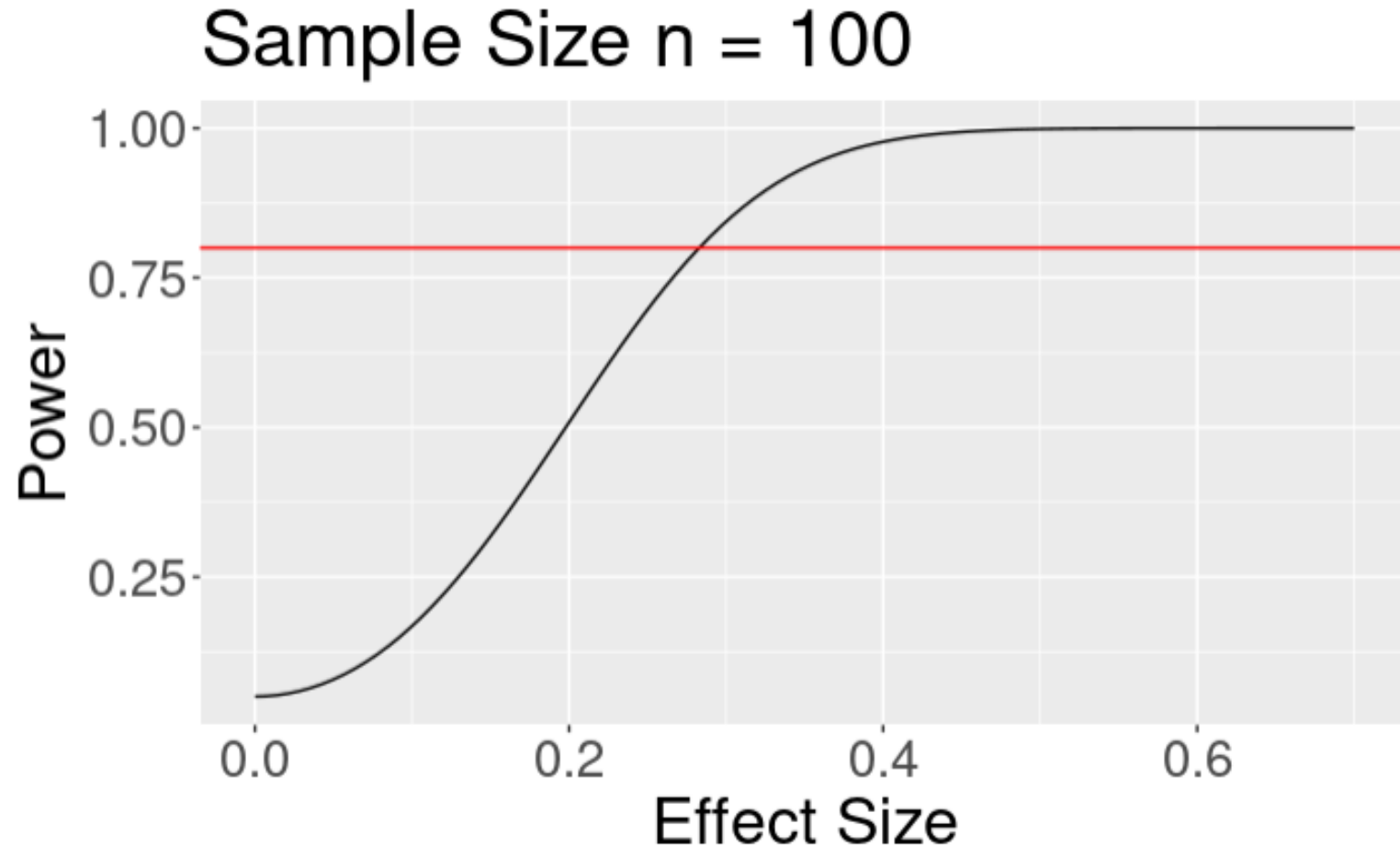
Example 1: Evaluate power along effect sizes with fixed sample size

```
#### Evaluate Power
```{r}
n_vec = 100
ES_vec = seq(0, 0.7, length.out = 100)
power_vec = rep(NA, 100)

d is the Effect Size
for(i in 1:100){
 # Find desired sample size
 power_vec[i] = pwr.t.test(n = 100, d = ES_vec[i],
 sig.level = 0.05,
 type = "one.sample", alternative =
"two.sided")$power
}

qplot(ES_vec, power_vec, geom = "line") +
 labs(x = "Effect Size", y = "Power", title = "Sample Size n = 100") +
 geom_hline(yintercept = 0.8, colour = "red")
```
```

Example 1: Evaluate power along effect sizes with fixed sample size



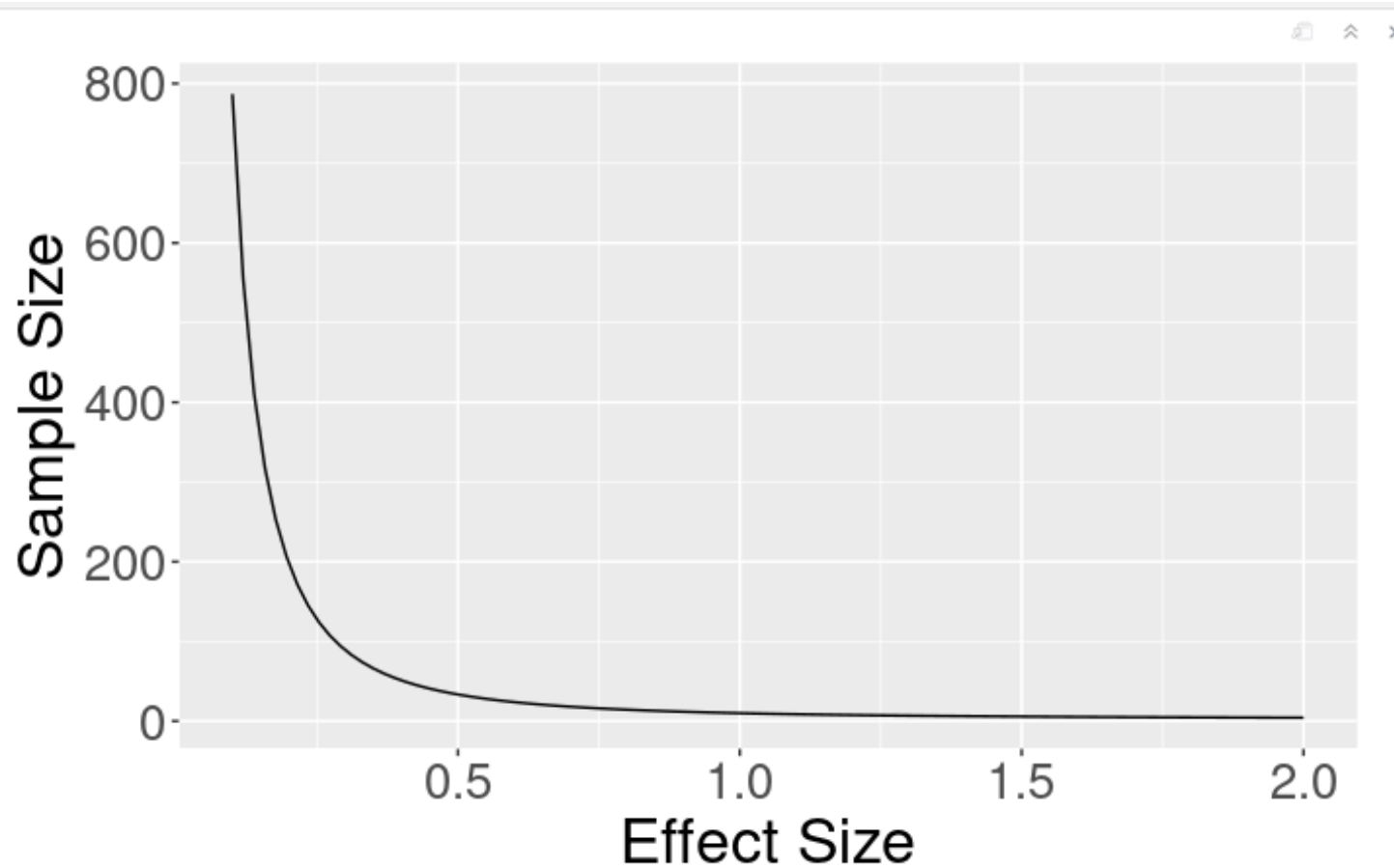
Example 1: Determine sample size for targeted 80% power

```
### Find desired sample size
```{r}
ES_vec = seq(0.1, 2, length.out = 100)
n_vec = rep(NA, 100)

d is the Effect Size
for(i in 1:100){
 # Find desired sample size
 n_vec[i] = pwr.t.test(d = ES_vec[i], sig.level = 0.05, power = 0.8,
 type = "one.sample", alternative = "two.sided")$n
}

qplot(ES_vec, n_vec, geom = "line") + labs(x = "Effect Size", y = "Sample
Size")
```
```

Example 1: Determine sample size for targeted 80% power



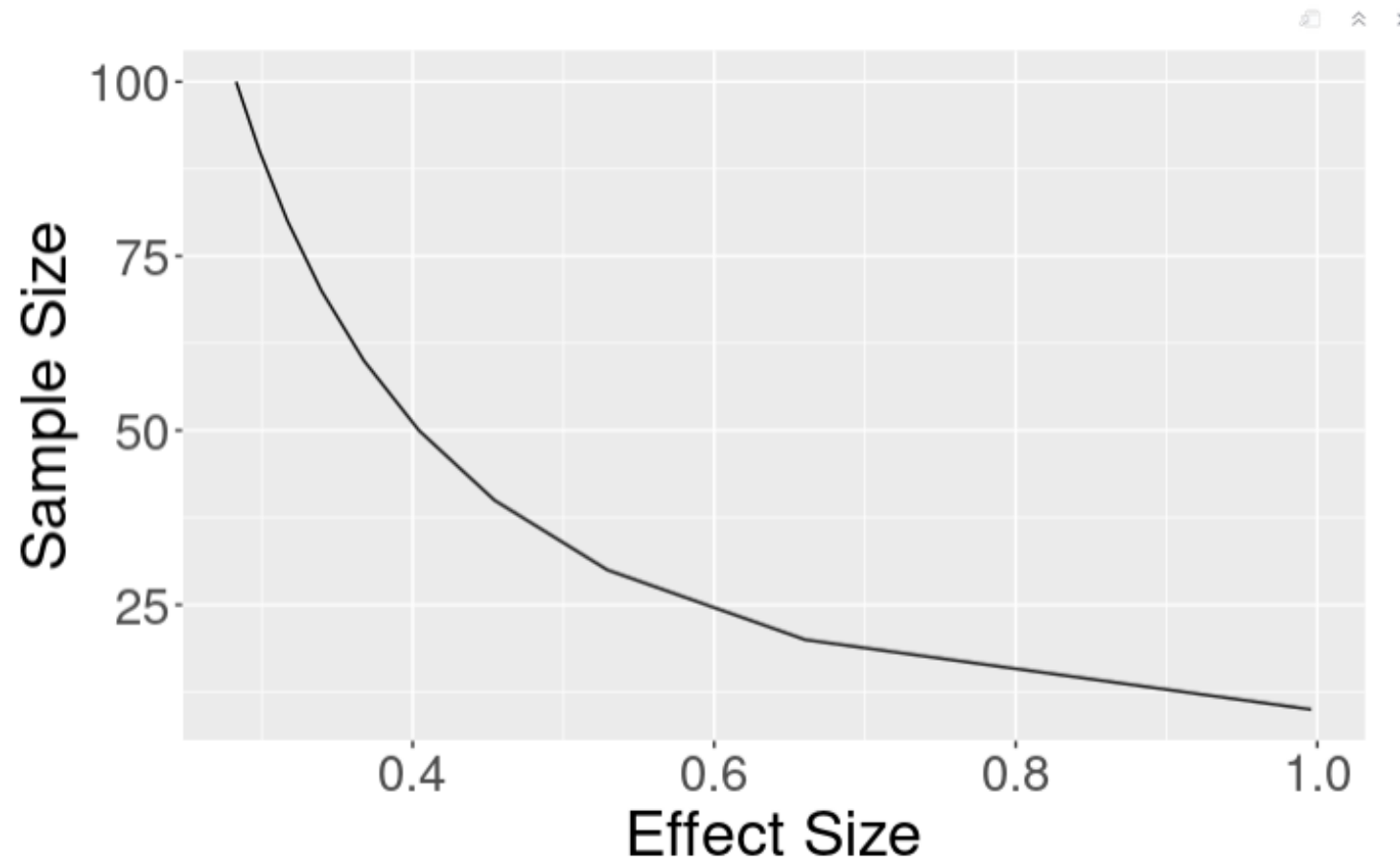
Example 1: Determine effect size for targeted 80% power

```
### Find desired effect size
```{r}
n_vec = seq(10, 100, by = 10)
ES_vec = rep(NA, 10)

d is the Effect Size
for(i in 1:10){
 # Find desired sample size
 ES_vec[i] = pwr.t.test(n = n_vec[i], sig.level = 0.05, power = 0.8,
 type = "one.sample", alternative = "two.sided")$d
}

qplot(ES_vec, n_vec, geom = "line") + labs(x = "Effect Size", y = "Sample
Size")
```
```


Example 1: Determine effect size for targeted 80% power



Example 2: Balanced one-way ANOVA

`pwr.anova.test`

Power calculations for balanced one-way analysis of variance tests

Description

Compute power of test or determine parameters to obtain target power (same as `power.anova.test`).

Usage

```
pwr.anova.test(k = NULL, n = NULL, f = NULL, sig.level = 0.05, power = NULL)
```

Arguments

| | |
|------------------------|---|
| <code>k</code> | Number of groups |
| <code>n</code> | Number of observations (per group) |
| <code>f</code> | Effect size |
| <code>sig.level</code> | Significance level (Type I error probability) |
| <code>power</code> | Power of test (1 minus Type II error probability) |

For a one-way ANOVA effect size is measured by f where

$$f = \sqrt{\frac{\sum_{i=1}^k p_i * (\mu_i - \mu)^2}{\sigma^2}}$$

where $p_i = n_i / N$,
 n_i = number of observations in group i
 N = total number of observations
 μ_i = mean of group i
 μ = grand mean
 σ^2 = error variance within groups

Cohen suggests that f values of 0.1, 0.25, and 0.4 represent small, medium, and large effect sizes respectively.

Example 2: Balanced one-way ANOVA

- Additional reading about how to determine sample size for functions in the R library "pwr":
 - https://bookdown.org/pdr_higgins/rmrwr/sample-size-calculations-with-pwr.html

Details

Exactly one of the parameters 'k', 'n', 'f', 'power' and 'sig.level' must be passed as NULL, and that parameter is determined from the others. Notice that the last one has non-NULL default so NULL must be explicitly passed if you want to compute it.

Value

Object of class "power.htest", a list of the arguments (including the computed one) augmented with 'method' and 'note' elements.

Example 2: Determine power with effect size 0.28, 4 groups, 20 samples per group

```
```{r}
Determine Power for given effect size f=0.28, 4 groups, 20 samples per group
pwr.anova.test(f=0.28, k=4, n=20, sig.level=0.05)
```
```

Balanced one-way analysis of variance power calculation

```
      k = 4
      n = 20
      f = 0.28
sig.level = 0.05
power = 0.5149793
```

NOTE: n is number in each group

Example 2: Determine sample size with effect size 0.28, 4 groups, targeted power 80%

```
```{r}
Determine required sample size for given effect size f=0.28, 4 groups, power = 0.8
pwr.anova.test(f=0.28, k=4, power=0.80, sig.level=0.05)
```
```

Balanced one-way analysis of variance power calculation

```
      k = 4
      n = 35.75789
      f = 0.28
sig.level = 0.05
power = 0.8
```

NOTE: n is number in each group

Experiment Design (Power Analysis)

- Choose appropriate test statistic/method
- Choose significance threshold: type I error tolerance rate ($\alpha = 0.05?$)
- Decide expected effect sizes based on preliminary data or previous results or pilot studies
- Plot power function vs. sample size, grouped by different effect sizes
- Goal
 - $\alpha \leq 0.05$ and sample size to achieve 80% power ($\beta=20\%$) in biomedical research
 - $\alpha \leq 0.05$ and sample size to achieve 90% power ($\beta=10\%$) in clinical trials
 - Lower significance threshold is needed when multiple testing exists

In-Class Credit

Complete Task 1.2 in the In-Class_Exercise.Rmd file for in-class credit.

Plot power curve vs. sample sizes with effect size 0.3.

Submit the html file of this exercise with your answers for Task 1.2 on canvas for in-class credit.

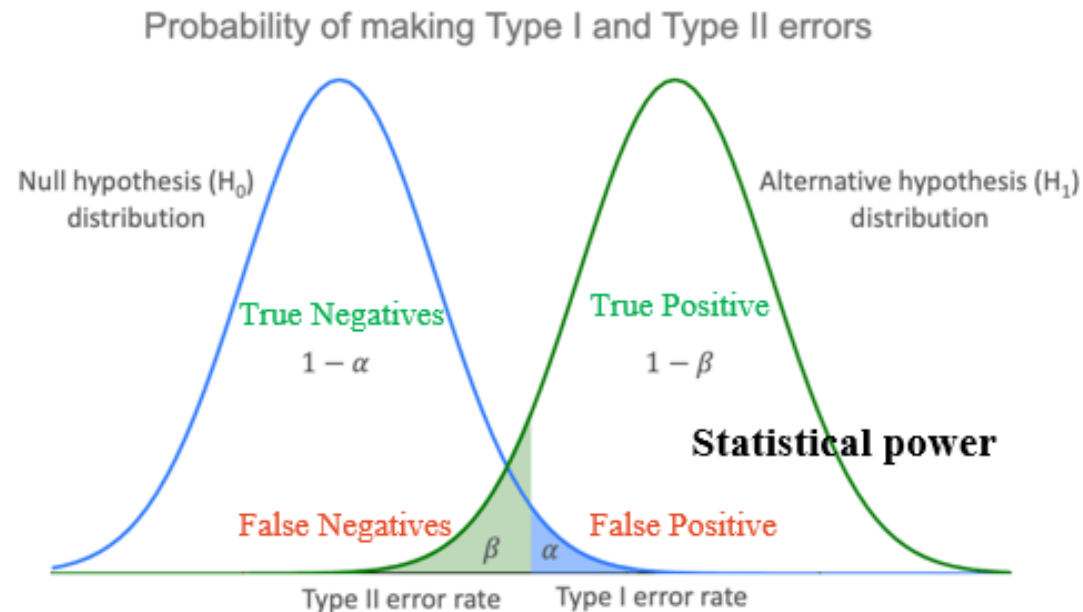
When Power Function is not Analytically Known

Monte Carlo Simulations

Simulation can also be used to evaluate type I error

Type I and Type II Error

| | H_0 is TRUE | H_0 is FALSE |
|------------------|---|---|
| Not Reject H_0 | Correct Inference
(true negative)
(probability = $1 - \alpha$) | Type II error
(false negative)
(probability = β) |
| Reject H_0 | Type I error
(false positive)
(probability = α) | Correct Inference
(true positive/ Power)
(probability = $1 - \beta$) |



Evaluate Power by Simulation

Simulate

Simulate outcome data under the Alternative hypothesis

- Calculate test statistic value
- Calculate p-value, and decide whether to reject the current simulation

Repeat

Repeat such simulation for a large number of iterations, e.g., 1000

Calculate

Calculate the proportion of simulations (POWER) that you indeed reject the null hypothesis



What would be
your desired
Power?

Evaluate Power by Simulation

With sample size
100, effect size 0.29

```
### Simulation to evaluate power for effect size = 0.29
```{r}
m = 1000 # number of simulation
n = 100 # sample size
sig_m = 0
alpha = 0.05
ES = 0.29
set.seed(123)
Simulate x variable values for m times, conduct t.test per simulation
for (i in 1:m){
 x = rnorm(n, mean = ES, sd = 1) # generate x under Ha
 x_mean = mean(x) # sample mean
 x_sd = sd(x) # sample deviation
 t_stat = x_mean / (x_sd / sqrt(n)) # t-test statistic value
 # Calculate p-value
 pvalue_t_test = pt(abs(t_stat), df = n - 1, lower.tail = FALSE) * 2
 if(pvalue_t_test < alpha){
 sig_m = sig_m + 1 # if significant
 }
}
Power
print(sig_m / m)
```
```

```
[1] 0.844
```

Evaluate Type I Error by Simulation

- Simulate outcome data under the NULL hypothesis
 - Calculate test statistic value
 - Calculate p-value
 - Decide whether to reject the current simulation
- Repeat such simulation for a large number of times
 - The total number of simulations will need to be determined based on the significance level you use
 - e.g., 1000 times with $\alpha = 0.05$, 10000 times with $\alpha = 0.005$
- Calculate the proportion of simulations (Type I Error) that you indeed reject the null hypothesis



What would be
your desired Type
I Error?

Evaluate Type I Error by Simulation

With sample size 100,
effect size 0

```
### Simulation to evaluate type I error
```{r}
m = 1000 # number of simulation
n = 100 # sample size
sig_m = 0
alpha = 0.05

set.seed(123)
Simulate x variable values for m times, conduct t.test per simulation
for (i in 1:m){
 x = rnorm(n, mean = 0, sd = 1) # generate x under NULL
 x_mean = mean(x) # sample mean
 x_sd = sd(x) # sample deviation
 t_stat = x_mean / (x_sd / sqrt(n)) # t-test statistic value
 # Calculate p-value
 pvalue_t_test = pt(abs(t_stat), df = n - 1, lower.tail = FALSE) * 2
 if(pvalue_t_test < alpha){
 sig_m = sig_m + 1 # if significant
 }
}

Type I Error
print(sig_m / m)
```
```

```
[1] 0.032
```

Next Lectures

- Week 12 (11/13): Permutation test (Jingjing)
 - Homework 7 due 11/13, revision due 11/20
- Week 13 (11/20): Machine learning (Jingjing)
- Week 14 (12/01): Adjust for multiple tests (Karen)
 - Homework 8 distributed and due 12/04
- Week 15 (12/08): Rigor and reproducibility (Karen)
 - Homework 8 revision due 12/11